Please attach this cover sheet to your completed homework

Homework 3 (Ch. 2) **ECE 311 – Hang Gao**

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Problem	Course outcome	Grade
2.7	1.c.	/10
2.8	1.c.	/10
2.11	1.c.	/10
2.16	1.c.	/10
2.17	1.c.	/10
2.19	1.c.	/10
2.24	1.c.	/10
2.25	1.c.	/10
2.26	1.c.	/10
2.34	1.c.	/10
	TOTAL:	/100

ABET Assessed Outcomes
1.c. Use appropriate models to formulate solutions related to electric circuit problems.

Homework #3
Due date: Sunday, February 15th by 11:59 PM (Electronically – Canvas)

2.7 Why are insulators used on power line towers?
2.8 Why are tower insulators built as disk shapes?
2.11 Why are transmission line towers higher than distribution line towers?
2.16 How can corona be reduced?
2.17 Why do we bundle conductors?
2.19 What are the roles of shield wires?
2.24 What is the main function of the CB?
2.25 What is the tap setting of the CB?
2.26 What is the time-dial setting of the CB?
2.34 What is the function of the surge arrester?

2.7 Since high-voltage towers are constructed using galvanized steel, which is electrically conductive, non-conductive insulators are placed between the transmission lines and the tower's frame to prevent electrifying the tower itself and creating a line-to-ground fault, or harming anyone who comes in contact with the tower.

2.8 When analyzing the potential for current to "jump" from the transmission line to the tower, bypassing the insulating material, the relevant variable is the creepage distance, which is defined as the shortest direct path along the surface of the insulator. The use of disks whose surface is continous with the rest of the insulators surface greatly increases the creepage distance while minimizing the distance at which the transmission line is held from the tower. This approach maximizes the insulators ability to prevent flashover along with structural integrity.

2.11 Due to the substantially lower voltage on distribution transmission lines, the flashover distance between the line and ground is reduced, and the towers can thus be lower to the ground in distribution networks.

2.16 The high electric field intesity induced by the high voltage on transmission lines can ionize the surrounding air. Essentially, this field accelerates free electrons in the air, generating current in the area immediately surrounding the tranmsission line, resulting in a number of undesireable effects. The associate leakage current is called **corona**, and it can be mitigated by increasing the diameter of the conductor, thus spreading its surface charge over a larger area.

2.17 Rather than constructing a single transmission line with a large diameter, however, several lines are often bundled to create a larger diameter while retaining lower production costs.

2.19 Lightning strikes can create undesireable electric surges if they impact transmission lines. To avoid this, shield wires are installed, which are conductive and connected to ground at every tower and substation. Since lightning will most often strike the highest point at the lowest potential, shield wires attract lightning strikes and safely conduct the resulting current to ground, protecting the transmission lines from unwanted overcurrent.

2.24 CB, which stands for circuit breaker, is a device that is normally installed at electrical substations for the purpose of breaking the electrically conductive path between a high-voltage transmission network and the substation's transformer. This device can be operated manually, or it can activate automatically when pre-determined current limits along its path are exceed.

2.25 The tap setting of a CB determines the maximum current limits, in excess of which the CB will open and break the electrically conductive path over which current flows from the transmission network to the transformer. The time-dial setting of the CB determines the amount of time that the current must remain in excess of the limit before the CB will open, since some faults are transient and can be cleared quickly without undesireable consequences.